

Radiotherapy for secondary bone cancer

You may have radiotherapy as part of your treatment for secondary bone cancer. This uses high-energy rays to destroy cancer cells in the bone.

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About radiotherapy for secondary bone cancer

Radiotherapy is often used to treat [secondary bone cancer](#). It can help relieve symptoms such as [pain](#).

Radiotherapy uses high-energy rays to treat cancer. It destroys cancer cells in the area where the radiotherapy is given.

Normal cells in the area can also be damaged by radiotherapy, but they can usually recover.

There are different ways of having radiotherapy. You may have external beam radiotherapy or radioisotope therapy.

External beam radiotherapy

[External beam radiotherapy](#) is given from a radiotherapy machine. You may have radiotherapy for secondary bone cancer as:

- a single treatment session
- a number of daily treatment sessions.

You usually have radiotherapy to the area of bone affected by the cancer. It may take 7 to 10 days for bone pain to start improving. It may take up to 6 weeks before you feel the full benefit of your treatment. Sometimes bone pain gets slightly worse before getting better.

It is important to continue taking any painkillers your healthcare professional prescribes. You can talk to your doctor or nurse about reducing your [painkillers](#) as your pain improves.

External radiotherapy does not make you radioactive. It is perfectly safe to be with other people after treatment, including children.

Secondary bone cancer forum

Join our online community forum to chat with other people who have been affected by secondary bone cancer, share your experiences, and ask questions.

Side effects of external radiotherapy

This type of radiotherapy usually causes very few [side effects](#). Side effects will depend on where the treatment is given.

Treatment to bones such as the ribs or the spine may make you feel sick for a couple of days afterwards. Some people may feel more tired or have some skin redness in the area where they have their treatment. Radiotherapy to your pelvis may also cause diarrhoea.

Your doctor, nurse or radiographer will discuss any possible side effects with you before your treatment starts. They will make sure you have any medication you need to manage side effects.

We have more general information about [radiotherapy](#).

Stereotactic ablative radiotherapy

[Stereotactic ablative radiotherapy](#) uses small beams of radiation to target the tumour. It is occasionally used to treat secondary bone cancer in the spine if only 1 or 2 areas are affected.

If your doctor thinks it is a suitable treatment for you they will give you more information about it.

Radioisotope therapy

[Radioisotopes](#) are radioactive substances. They may be given as a drink, capsules or an injection into a vein. Cancer cells absorb radioisotopes more than normal cells and get a higher dose of radiation.

All the bones affected by cancer will be treated. It can be a helpful treatment if several of your bones are affected.

There are different types of radioisotopes. The type you have will depend on the type of primary cancer you have:

- Radium-223 can be used to treat secondary cancer in the bone from [prostate cancer](#) that is no longer responding to hormonal therapy.

- Strontium-89 can be used to treat secondaries in the bone from prostate cancer and [breast cancer](#).
- Iodine-131 can be used to treat secondaries in the bone from [thyroid cancer](#).

Your cancer doctor will discuss your treatment with you and answer any questions you might have.

Side effects of radioisotope treatment

Radioisotopes such as radium-223 or strontium-89 can temporarily reduce the number of normal red and white blood cells produced by the bone marrow.

Your doctor will arrange [blood tests](#) before your treatment to check the level of your blood cells. If the number of your red blood cells is low (anaemia), you may feel tired and breathless. If they are very low, your doctors may give you a blood transfusion.

When the number of white blood cells is low you are more likely to get an [infection](#). If you do, your doctors will prescribe antibiotics.

Some people have swelling around the tumour area in the days following treatment. This is called tumour flare and can cause a temporary increase in [pain](#) and tenderness. You may need to take [painkillers](#) for a few days.

About our information

References



Below is a sample of the sources used in our secondary bone cancer information. If you would like more information about the sources we use, please contact us at cancerinformationteam@macmillan.org.uk

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Reviewers



This information has been written, revised and edited by Macmillan Cancer Support's Cancer Information Development team. It has been reviewed by expert medical and health professionals and people living with cancer. It has been approved by Chief Medical Editor, [Professor Tim Iveson](#), Consultant Medical Oncologist.

Our cancer information has been awarded the [PIF TICK](#). Created by the Patient Information Forum, this quality mark shows we meet PIF's 10 criteria for trustworthy health information.

Date reviewed

Reviewed: 01 February 2023 | Next review: 01 February 2026

ⓘ This content is currently being reviewed. New information will be coming soon.

Our cancer information meets the [PIF TICK](#) quality mark.

This means it is easy to use, up-to-date and based on the latest evidence. Learn more about [how we produce our information](#).

How we can help

Macmillan Support Line

The Macmillan Support Line is a free and confidential service for people living with and affected by cancer. If you need to talk, we'll listen.



Call us free on 0808 808 0000

8am to 8pm, 7 days a week. You can talk to our cancer specialists about anything.



Email the Macmillan Support Line

Email the Macmillan Support Team if you have any questions.

Chat online

Available 8am to 7.45pm, 7 days a week. For confidential, real-time support if you don't want to call.



Online Community

Chat online anonymously to others who understand what you are going through. Our community is available 24/7 and has dedicated forums where you can get advice and ask our experts.



Find cancer support services in your area

What's going on near you? Find out about support groups, where to get information and how to get involved with Macmillan where you live.

[Get help](#)